SPX

DRAG-TYPE UNDERREAMER, PDC CUTTERS

ENLARGEMENT

GENERAL DESCRIPTION

DRAG-TYPE UNDERREAMER WITH SPX PDC CUTTERS ON THE CUTTING EDGE: THE HARDNESS AND WEAR RESISTANCE OF POLYCRYSTALLINE DIAMOND, BACKED BY THE STRENGTH OF CEMENTED TUNGSTEN CARBIDE. THE CUTTERS SHEAR THROUGH SOFT TO MEDIUM-HARD FORMATIONS FASTER THAN CONVENTIONAL TOOLS AND NEED LESS TOOL WEIGHT. USED IN POORLY CONSOLIDATED SOFT TO MEDIUM FORMATIONS WHERE LARGER DIAMETERS ARE REQUIRED FOR GRAVEL PACKING OR CEMENTING. A PILOT HOLE CAN BE ENLARGED UP TO THREE TIMES THE BODY DIAMETER IN A SINGLE TRIP.

TECHNICAL CHARACTERISTICS

Tool series	Body Ø	Min. pilot hole Ø	Std. expanded Ø, SPX (PDC)	Std. expanded Ø, tungsten carbide	Fishing neck length	Fishing neck Ø	Overall length	Top pin API	Wt (lb)
3600	3-3/8	3-3/4	N/A	9	8	3-3/8	26-1/2	2-3/8 Reg.	tbc
4500	4-1/2	4-3/4	6-1/2, 6-3/4, 8-1/2	Upon request	18	4-1/4	69	3-1/2 IF	230
4700	4-3/4	5	Upon request	12	18	4-1/8	67	2-7/8 Reg.	tbc
5700	5-3/4	6	7-1/2, 8, 8-1/2, 12, 13	16	18	4-3/4	70	3-1/2 Reg.	350
7200	7-1/4	7-5/8	10, 12-1/4, 13, 14, 15, 16	22	18	5-3/4	78	4-1/2 Reg.	750
8200	8-1/4	8-1/2	10, 12-1/4, 14, 15, 16, 17	23	18	5-3/4 or 8	78	4-1/2 or 6-5/8 Reg.	tbc
9500	9-1/2	9-7/8	12-1/4, 17-1/2	28	18	8	78	6-5/8 Reg.	tbc
11700	11-3/4	12-1/4	Upon request	36	18	8 or 9	86	6-5/8 or 7-5/8 Reg.	tbc

All dimensions in inches, weights in lb (approximate). Other expanded diameters, orifices and connections available on request. tbc: to be confirmed.

DESIGN FEATURES

- **Self-cleaning arms:** a circulation jet nozzle sends part of the flow out of each arm pocket, cleaning the cutting surfaces, improving cuttings removal and dissipating frictional heat.
- **Durable arms:** dressed with long-wearing cutting-grade tungsten carbide.
- **Field serviceable:** low-cost cutter arms and orifices can be replaced in the field.
- **Optional Flo-Tel:** pump pressure drops by 200 - 250 psi when the arms are fully extended, confirming full-gauge underreaming.

PERFORMANCE PARAMETERS

Rotary speed, cutting the shoulder	40 - 60 rpm
Flow rate	Increase gradually, 250 - 450 gpm
Weight on tool	Start at 3,000 lb, up to 10,000 lb
With Flo-Tel	80 - 150 rpm at max. pump pressure
Circulating medium	Water, mud, air, aerated mud, foam
Max. enlargement	3 x body Ø in a single trip

ADVANTAGES

- Faster penetration and more time on bottom
- Less rig time and lower total cost per foot
- Quick drilling through producing zones limits formation damage and hole-stability problems

HOW TO ORDER

Please specify: 1. tool series 2. top and bottom connections 3. fishing neck Ø 4. opening Ø 5. size and weight of casing to run through, if known 6. bullnose or bits (ordered separately) 7. cutting structure (milled tooth, TCI or PDC).

